

### Cel ćwiczenia

Zbudowanie prostej aplikacji webowej z formularzem w Thymeleaf (z walidacją i PRG - Post-Redirect-Get).

### Niezbędne narzędzia

1. IDE: dowolne (STS 4 / IntelliJ / VS Code).
2. Maven: 3.9+ (albo wrapper z Initializr).
3. JDK: 21.

### Generowanie projektu (Spring Initializr)

1. Project: Maven
2. Language: Java
3. Spring Boot: 3.3.x
4. Java: 21
5. Dependencies:
  - Spring Web
  - Thymeleaf
  - Validation (Jakarta Validation)
  - Lombok

### Struktura projektu

```
src/
├── main/
│   ├── java/
│   │   ├── edu/
│   │   │   ├── zut/
│   │   │   │   ├── awir/
│   │   │   │   │   ├── BootApplication.java
│   │   │   │   │   ├── controller/
│   │   │   │   │   │   ├── UserController.java
│   │   │   │   │   └── model/
│   │   │   │   │       └── User.java
│   └── resources/
│       ├── templates/
│       │   ├── add-user.html
│       │   └── user-info.html
│       └── application.properties
```

### Zawartość pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.
0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
```

```

    <version>3.3.4</version>
    <relativePath/>
</parent>

<groupId>edu.zut.awir</groupId>
<artifactId>awir2</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>AWiR Ćwiczenie 2</name>

<properties>
    <java.version>21</java.version>
</properties>

<dependencies>
    <!-- Spring Boot -->
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-thymeleaf</artifactId>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-validation</artifactId>
    </dependency>

    <!-- Lombok -->
    <dependency>
        <groupId>org.projectlombok</groupId>
        <artifactId>lombok</artifactId>
        <optional>true</optional>
    </dependency>
</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>

```

### **Klasa domenowa**

```

package edu.zut.awir.model;

import jakarta.validation.constraints.Email;

```

```
import jakarta.validation.constraints.Min;
import jakarta.validation.constraints.NotBlank;
import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;
```

```
@Data
@NoArgsConstructor
@AllArgsConstructor
public class User {
```

```
    @Min(1)
    private long id;
```

```
    @NotBlank(message = "Name is required")
    private String name;
```

```
    @NotBlank(message = "Email is required")
    @Email(message = "Email must be valid")
    private String email;
```

```
}
```

## Kontroler

```
package edu.zut.awir.controller;
```

```
import edu.zut.awir.model.User;
import jakarta.validation.Valid;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.validation.BindingResult;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestMapping;
import
org.springframework.web.servlet.mvc.support.RedirectAttributes;
```

```
@Controller
@RequestMapping("/users")
public class UserController {
```

```
    @GetMapping("/new")
    public String showForm(Model model) {
        if (!model.containsAttribute("user")) {
            model.addAttribute("user", new User());
        }
        return "add-user";
    }
```

```
    @PostMapping
    public String processForm(@Valid @ModelAttribute("user") User
user,
```

```

        BindingResult result,
        RedirectAttributes redirectAttributes)
{
    if (result.hasErrors()) {
        redirectAttributes.addFlashAttribute("org.springframework.validation.BindingResult.user", result);
        redirectAttributes.addFlashAttribute("user", user);
        return "redirect:/users/new";
    }
    redirectAttributes.addFlashAttribute("user", user);
    return "redirect:/users/info";
}

@GetMapping("/info")
public String showInfo() {
    return "user-info";
}
}

```

## Widoki

src/main/resources/templates/add-user.html

```

<!DOCTYPE html>
<html lang="pl" xmlns:th="http://www.thymeleaf.org">
<head>
    <meta charset="UTF-8">
    <title>Rejestracja użytkownika</title>
    <style>
        label { display:block; margin-top:.5rem; }
        .error { color:#b00020; font-size:.9rem; }
        .field { margin-bottom:.75rem; }
    </style>
</head>
<body>
<h1>Formularz rejestracji</h1>

<form th:action="@{/users}" th:object="${user}" method="post">
    <div class="field">
        <label for="id">Id:</label>
        <input id="id" type="number" th:field="*{id}" />
        <div class="error" th:if="${#fields.hasErrors('id')}"
th:errors="*{id}"></div>
    </div>

    <div class="field">
        <label for="name">Name:</label>
        <input id="name" type="text" th:field="*{name}" />
        <div class="error" th:if="${#fields.hasErrors('name')}"
th:errors="*{name}"></div>
    </div>

    <div class="field">

```

```

        <label for="email">Email:</label>
        <input id="email" type="email" th:field="*{email}" />
        <div class="error" th:if="${#fields.hasErrors('email')}"
th:errors="*{email}"></div>
    </div>

    <p>
        <input type="submit" value="Submit">
        <input type="reset" value="Reset">
    </p>
</form>

</body>
</html>

```

src/main/resources/templates/user-info.html

```

<!DOCTYPE html>
<html lang="pl" xmlns:th="http://www.thymeleaf.org">
<head>
    <meta charset="UTF-8">
    <title>Informacja o użytkowniku</title>
</head>
<body>
<h1>Użytkownik</h1>

<p th:if="${user == null}">
    Brak danych. <a th:href="@{/users/new}">Wróć do formularza</a>
</p>

<div th:if="${user != null}">
    <p th:text="'id: ' + ${user.id}"></p>
    <p th:text="'name: ' + ${user.name}"></p>
    <p th:text="'email: ' + ${user.email}"></p>
    <a th:href="@{/users/new}">Kolejny użytkownik</a>
</div>

</body>
</html>

```

## Uruchomienie aplikacji

Po uruchomieniu aplikacji należy z przeglądarki wywołać adres:  
<http://localhost:8080/users/new>